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Steel Surface Defect Detection Based on an Improved YOLO26 Algorithm

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***Abstract*—Targeting the industrial challenges of limited edge computing resources on-site in the steel industry, low contrast of surface defects on hot-rolled steel strips, and the tendency for small irregular defects to be missed during traditional downsampling due to information loss, this paper proposes an improved algorithm model (YOLO26-SPD-CMSA) based on the lightweight network YOLO26, which integrates spatial lossless downsampling with a contextual multi-scale spatial attention mechanism. First, a Spatial-to-Depth conversion module (SPD) is introduced to reconstruct the backbone network, replacing traditional strided convolutions to avoid the loss of fine target information. Subsequently, to address the issues of spatial positional ambiguity and difficulty in handling multi-scale defects introduced by SPD, a Contextual Multi-scale Spatial Attention mechanism (CMSA) is designed. This mechanism organically combines the bidirectional one-dimensional feature spatial aggregation capability of Coordinate Attention (CA) with the receptive field advantages of Multi-Scale Channel Attention (MSCA), achieving precise spatial localization, denoising, and multi-scale adaptivity of features. All evaluation metrics are based on the average of three independent trainings, along with standard deviation statistics. In validations on the public NEU-DET and GC10-DET steel defect datasets, YOLO26-SPD-CMSA (with 2.94M parameters and 8.7 GFLOPs) achieved mAP50 scores of 75.22% and 65.50%, respectively, which are improvements of 1.36% and 2.36% over the baseline. Compared with current mainstream lightweight networks and some heavyweight ones, it provides a very valuable lightweight solution for industrial production lines to capture surface defects in real time even with limited edge computing capabilities.**

Keywords*: *Steel surface defect detection; YOLO26; Contextual Multi-Scale Spatial Attention (CMSA); Lightweight Network

I. INTRODUCTION

As a core material in modern industry, construction, and manufacturing, the surface quality of steel directly affects the reliability and safety of industrial products. In the production process of hot-rolled steel strips, due to the complex production environment and process errors, steel surfaces inevitably develop various types of defects such as cracks, inclusions, pitted surfaces, and scratches[1]. Traditional manual visual inspection is not only inefficient and labor-intensive but also prone to misdetection due to eye fatigue and inconsistent subjective judgment standards, making it inadequate to meet the requirements for high-quality steel in modern industry[2]. Therefore, against the backdrop of Industry 4.0 and intelligent manufacturing, the real-time detection of steel surface defects through the integration of computer vision and deep learning has become an urgent problem that needs to be addressed[3].

In recent years, deep learning-based object detection algorithms have made breakthrough progress in the field of industrial vision. As the culmination of real-time object detection, the YOLO network proposed by Redmon et al.[4] broke the limitations of traditional two-stage detection and transformed the object detection problem into a regression problem of spatially separated bounding boxes and relevant class probabilities, laying a solid foundation for efficient real-time detection. Subsequently, the YOLO series underwent long-term iterations and co-evolution. Ge et al. [5] proposed YOLOX, which further improved the network's generalization and detection capabilities by introducing a decoupled head and an efficient label assignment strategy, SimOTA; Wang et al.[6] proposed YOLOv7, which achieved a good balance between speed and accuracy by introducing a trainable Bag-of-freebies strategy; Reis et al. [7] based on YOLOv8, further optimized the feature extraction network, demonstrating strong transfer learning capabilities; Wang et al. [8] based on YOLOv9 proposed the Programmable Gradient Information (PGI) theory to address the various changes required for deep networks to handle multiple targets. Recently, Wang et al. [9] proposed YOLOv10, which introduced an NMS-Free integrated end-to-end strategy, significantly reducing inference latency while setting a new record for detection efficiency. Tian et al.[10], based on YOLOv12, for the first time proposed a real-time object detection framework centered on the attention mechanism, breaking its speed bottleneck while fully utilizing the attention mechanism to improve detection accuracy and maintaining extreme inference speed comparable to traditional CNN architectures.

With the iterative improvement of algorithms, lightweight YOLO26 variants with high spatial localization accuracy have emerged for industrial scenarios and clinical detection tasks [11]. PADMANABHA et al.[12]ound through experiments that YOLO26 variants can achieve better accuracy and efficiency in resource-constrained environments at very low computational cost when facing high-precision detection and specific industrial component localization. However, when applying the basic YOLO26 algorithm to detect the classic steel surface defect dataset NEU-DET, the model still faces challenges.

For the aforementioned industrial scenarios, this study proposes an improved YOLO26-SPD-CMSA algorithm based on the YOLO26 algorithm, featuring lossless feature

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preservation and multidimensional collaborative perception. First, to address the issue of tiny feature breakage and information loss caused by traditional stride convolution operators, this paper reconstructs the backbone network by introducing a Spatial-to-Depth conversion (SPD) module [13]. This module avoids the loss of small object information through lossless slicing and channel multidimensional stacking mechanisms. However, the SPD module compresses two-dimensional spatial information into high-dimensional channels, which inevitably causes spatial position ambiguity and channel feature redundancy, leading to degradation in the network's scale adaptation to dense pitting and complex inclusions. To address this, a single-scale coordinate attention (CA) [14]was added, but it is difficult to simultaneously accommodate scale variations of defects. Therefore, in this paper, a Contextual Multi-Scale Spatial Attention mechanism (CMSA) is proposed after the SPD downsampling unit. This mechanism mitigates the dependence of multi-scale spatial perception on global average pooling, injects the one-dimensional directional position joint encoding flow of CA as the channel perception base to accurately locate defect coordinates, and simultaneously connects a bar-shaped multi-scale large-kernel perception matrix, which, through lossless interception, fits irregular microscopic morphologies. Through the coupling of SPD and CMSA, the model can detect complex defect features with relatively low parameters and computational cost.

II. The Improvement Method Proposed In This Paper

A. YOLO26-SPD-CMSA

To enhance the detection of tiny, irregularly shaped defects on hot-rolled steel strips while maintaining an ultra-lightweight model, this paper proposes the YOLO26-SPD-CMSA architecture, based on coordinate-aware multi-scale collaborative perception; the overall architecture is illustrated in Figure 1.

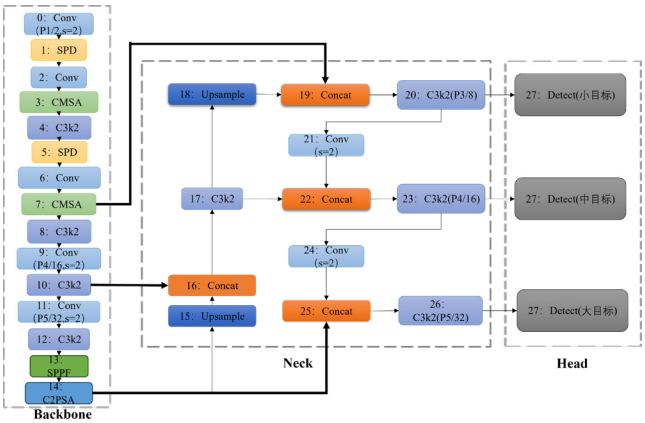


Figure 1. YOLO26-SPD-CMSA Architecture

The architectural design adheres to the principles of lossless preservation of multi-dimensional spatial details and flexible, position-sensitive scale calibration. During the shallow feature extraction phase of the backbone network, the downsampling operations caused by strided convolutions in the original network are replaced by a lossless downsampling unit comprising a Space-to-Depth (SPD) module and a standard convolution with a stride of 1. Through two successive space-to-depth remapping operations, high-dimensional channel information is enhanced without the abrupt loss of spatial resolution, resulting in multi-channel feature maps with strong representational power for low-frequency textures.

Subsequently, to address channel redundancy resulting from lossless downsampling and the limitations of traditional Coordinate Attention (CA) in adapting across scales for complex, irregularly shaped defects, a Contextual Multi-Scale Spatial Attention (CMSA) mechanism is introduced prior to the output of shallow-to-mid-level feature maps. This mechanism overcomes the issue where traditional multi-scale perception relies excessively on global average pooling—leading to a loss of spatial positional cues—by integrating the 1D directional positional encoding stream of CA as a foundation for channel-based defect localization. Simultaneously, a lightweight, direction-selective strip-based multi-scale large-kernel matrix is decoupled and implemented in a parallel spatial branch to precisely fit the shapes of irregular defects.

This dual-path parallel design achieves filtering of redundant high-dimensional channels and spatial noise reduction while maintaining a lightweight profile. Finally, these shallow, high-resolution features—possessing both precise directional sensitivity and multi-scale morphological alignment—are effectively transmitted to shallow, mid-level, and deep detection heads via the multi-scale concatenation operators of the feature pyramid (Neck), thereby enabling the effective fusion of global heterogeneous features in a pure training-from-scratch scenario.

B. Space-to-depth conversion mechanism

Strided convolutions or max-pooling operations in traditional CNNs discard neighboring pixels when reducing spatial resolution, which is highly detrimental for detecting fine cracks and minute pitting in images. Therefore, this paper introduces the Space-to-Depth (SPD) module[13]; as shown in Figure 2, it redistributes spatial information into the channel dimension without loss while simultaneously reducing resolution.

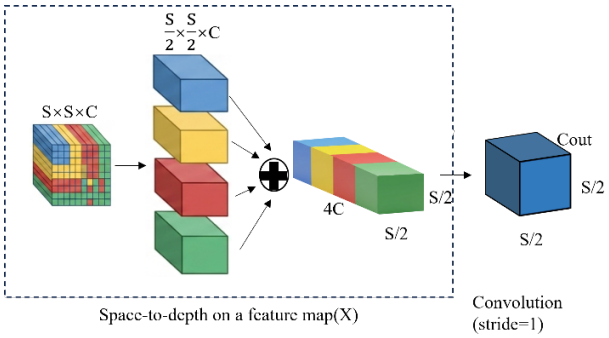


Figure 2. SPD module

Let the input 2D feature map be denoted by X, with dimensions S × S × C. The slicing and concatenation process of the SPD module can be described as follows: the original feature map is partitioned into four isotropic sub-feature maps—X0,0, X1,0, X0,1, and X1,1—based on specific strides and offsets. The mathematical mapping relationship for this slicing can be uniformly expressed as:

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$$X_{i,j} = X[i{:}S{:}2, j{:}S{:}2, :] \ \ (i,j \in \{0,1\}) \qquad (1)$$

Following this spatial discretization and extraction process, the spatial resolution of each of the four sub-feature maps is reduced to half of the original—meaning their dimensions become S/2 × S/2 × C. Subsequently, the SPD operator concatenates these four spatially complementary sub-feature maps along the channel dimension to produce the cascaded feature map XSPD:

$$X_{\mathrm{SPD}} = \mathrm{Concat}\left(\left[\mathrm{X}_{0,0}, \mathrm{X}_{1,0}, \mathrm{X}_{0,1}, \mathrm{X}_{1,1}\right]\right) \in \mathrm{R}^{\frac{\mathrm{S}}{2}\times\frac{\mathrm{S}}{2}\times 4\mathrm{C}} \qquad (2)$$

Through this mechanism, although the spatial area of the feature map is reduced fourfold, the low-frequency geometric details and edge continuity of micro defects are encoded into the four-fold expanded channel space. A subsequent stride-1 convolution fuses cross-channel features to eliminate redundant information and realize lossless spatial downsampling.

However, channel expansion causes extra memory bandwidth overhead. As quantified in Table 1, SPD only brings a 3.77% increase in weight memory, while the total memory access cost and peak feature memory rise moderately by 9.12% and 9.27%, respectively. The mild memory increment is an acceptable trade-off for retaining complete pixel-level defect details lost by standard pooling layers.

TABLE I. MEMORY OVERHEAD COMPARISON BETWEEN BASELINE AND SPD-EMBEDDED MODEL

Model	*Weight Memory (MB)*	*Total MACs (MB)*	*Peak Feature Memory (MB)*
Baseline	9.81	355.16	345.42
YOLO26-SPD	10.18	387.56	377.45
Growth Rate (%)	+3.77%	+9.12%	+9.27%

C. Coordinate self-attention mechanism

While the SPD module preserves low-frequency spatial details, it causes a rapid surge in the number of channels over a very short distance (reaching 64 and 256 channels after scaling). This makes the network highly susceptible to the introduction of background reflection noise and spatial misalignment within the non-linear layers. To address this, we introduce the Coordinate Attention (CA) mechanism to dynamically recalibrate the high-dimensional channel feature streams[14].

Traditional channel attention employs global 2D average pooling, which is prone to losing the coordinates of small objects. The CA module effectively mitigates this limitation by using a pair of 1D spatial pooling kernels to aggregate features along the horizontal (H) and vertical (W) coordinate directions, respectively.

The one-dimensional feature vectors generated along these two orthogonal directions possess powerful directional awareness capabilities. Subsequently, the module concatenates the generated horizontal and vertical feature vectors and feeds them into a shared 1×1 convolutional layer for non-linear mapping and channel compression, employing Batch Normalization (BN) and ReLU activation for feature denoising. Immediately following this, the feature map is split along the spatial dimension into two independent feature vectors; two additional 1×1 convolutional layers are used to restore the channel count to its original size, and finally, Sigmoid activation functions are applied to generate horizontal and vertical attention weight maps.

The design ensures that the dense information flow transferred from the SPD module is effectively filtered and denoised, with position-aware weights injected in both horizontal and vertical directions.

D. Multi-scale spatial geometric flow

Multi-Scale Convolutional Attention (MSCA) integrates multi-scale convolution with an attention mechanism[15]. By employing lightweight multi-scale fusion and dynamic attention allocation, it achieves dual optimization—enhancing features of small objects while suppressing complex noise—within a limited computational budget, thereby offering an effective solution for real-time detection in dynamic environments. Its core innovation lies in achieving multi-scale feature extraction and dynamic spatial attention through low-complexity operations.

To address the specific detection task of this study, we have modified the spatial geometric flow topology of the MSCA. For the input feature map, basic spatial features are first captured via a 1×1 convolution, accompanied by channel-level compression (C→1):

$$\mathrm{F}_{\mathrm{base}} = \mathrm{Conv}_{1\times1}(\mathrm{X}) \qquad (3)$$

E. Contextual Multi-Scale Spatial Attention Mechanism (CMSA)

To enhance the capability of detecting long, narrow, and irregularly shaped surface defects on hot-rolled steel strips, this paper designs the CMSA module, as illustrated in Figure 3. This module employs a dual-stream parallel weighted architecture.

Features along the H and W dimensions are aggregated via 1D global average pooling; after concatenation, the feature map undergoes dimensionality reduction using a 1×1 convolution, followed by Batch Normalization (BN) and ReLU activation. Subsequently, the features are split along the spatial dimension, and independent 1×1 convolutions combined with Sigmoid activation are used to restore the original channel count, generating direction-aware horizontal and vertical attention weights, Cah and Caw.

Parallel, lightweight, asymmetric strip convolution cascades (with kernel sizes K=5, 7, and 9) are utilized to extract multi-scale spatial morphological features. In each branch, the channel dimension is first compressed to 1, followed by 1×K and K×1 convolutions to capture the orientation and shape of irregular defects. Finally, the features from the various branches are averaged via pixel-wise summation and passed through a Sigmoid activation function to generate the spatial attention weight map, Sa.

Ultimately, the three sets of attention weights are applied to the original input feature map X through channel-wise and

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spatial broadcasting multiplication, collaboratively producing the calibrated feature map:

$$Y = X \times Ca_h \times Ca_w \times S_a \quad (4)$$

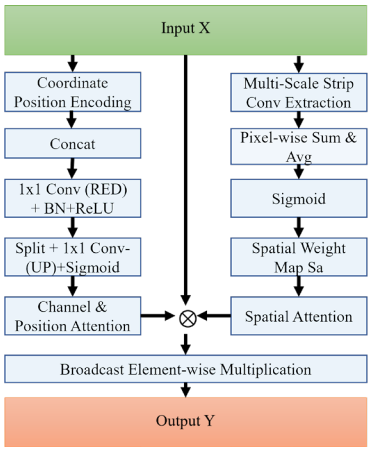


Figure 3. CMSA module

III. Experimental Results And Scientific Discussion

A. Datasets and Experimental Environment

To evaluate the performance of the proposed YOLO26-SPD-CMSA model, systematic testing was conducted on the publicly available NEU-DET steel surface defect dataset. This dataset comprises six common types of defects found on hot-rolled steel strips: crazing, inclusion, patches, pitted surface, rolled-in scale, and scratches. Each defect category contains 300 grayscale images (200×200 pixels), resulting in a total of 1,800 samples. For the experiments, the dataset was randomly partitioned into training, testing, and validation sets using a 7:2:1 ratio.

All comparative experiments and ablation studies in this research were performed on identical high-performance computing hardware, featuring an NVIDIA GeForce RTX 4090 GPU (24GB VRAM) and an Intel Xeon high-performance processor. The software environment was built upon the Ubuntu 20.04 operating system, the PyTorch 1.9.0 deep learning framework, and the CUDA 11.1 acceleration library. The model was trained for a total of 300 epochs with a uniform input image size of 640×640 pixels and a batch size of 16. AdamW was selected as the optimizer, with an initial learning rate of 0.01, a momentum factor of 0.937, and a weight decay coefficient of 0.001; a linear decay strategy was employed for the learning rate schedule (cos_lr=False), and pre-trained weights were disabled (pretrained=False). To mitigate errors arising from random weight initialization and non-deterministic floating-point calculations in the hardware, all algorithms in the comparative groups were trained independently three times under identical hardware and software conditions, and results are reported as Mean ± Standard Deviation (Mean±SD).

B. Evaluation metrics

To comprehensively and quantitatively evaluate the performance of the improved model proposed in this paper—YOLO26-SPD-CMSA—in the task of steel surface defect detection, standard evaluation metrics from the field of object detection are employed. These include Precision (P), Recall (R), Average Precision (AP), and mean Average Precision (mAP).

The definitions and calculation formulas for the metrics are as follows:

1) Precision (P): Measures the proportion of true positive samples among the bounding boxes predicted by the model as positive samples. Its calculation formula is:

$$P = \frac{TP}{TP+FP} \quad (5)$$

2) Recall (R): It measures the proportion of all actual positive samples in the dataset that are successfully predicted by the model. The formula for calculating it is:

$$R = \frac{TP}{TP+FN} \quad (6)$$

In the formulas, TP (True Positive) represents the number of detection boxes where the Intersection over Union (IoU) exceeds the set threshold and the class prediction is correct; FP (False Positive) represents the number of false detections—either background misidentified as a defect or detection boxes with an IoU below the set threshold; and FN (False Negative) represents the number of actual defect targets that failed to be detected. In this study, the IoU threshold is uniformly set to 0.5.

3) Mean Average Precision (mAP): AP represents the area under the P-R curve, reflecting the model's detection performance for a single defect category. mAP is the arithmetic mean of the AP values across all defect categories and serves as the primary metric for evaluating the overall performance of multi-class object detection. Its calculation formula is:

$$AP = \int_0^1 P(R)dR \quad (7)$$

$$mAP = \frac{1}{N}\sum_{i=1}^{N} AP_i \quad (8)$$

In the formula, N represents the total number of defect categories; for the NEU-DET dataset used in this study, N = 6. The mAP50 metric refers to the mean average precision calculated at an IoU threshold of 0.5, while mAP50-95 represents the aggregate mean average precision calculated as the IoU threshold increases from 0.5 to 0.95 in increments of 0.05.

C. Core Ablation Experiments and Statistical Analysis

Table 2 summarizes the controlled-variable ablation results of each core component. The performance variation of each model variant is analyzed as below:

Experiment 2 validates that SPD brings consistent improvement of detection metrics by retaining complete micro-defect edge details, and the mAP50-95 metric reflecting localization accuracy increases by 1.14%.

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To explore the synergistic effect of CA and MSCA, we carry out equal-parameter comparison experiments among single CA, single MSCA, classic CBAM and the proposed CMSA. Under nearly identical parameters and computational overhead:

Single CA (Exp.3) improves prediction stability but fails to raise the overall mAP50 compared with YOLO26-SPD;

Single MSCA (Exp.4) reduces missed detection but causes severe precision loss;

The mainstream CBAM attention (Exp.5) results in a sharp drop of mAP50 to 71.48%, showing poor adaptability to micro-defect detection tasks.

Benefiting from the complementary global-local feature extraction mechanism of CA and MSCA, the composite CMSA (Exp.6) achieves the highest mAP50 of 75.22% with stable performance, which fully verifies the 1+1>2 synergistic enhancement effect rather than parameter expansion.

TABLE II. COMPREHENSIVE ABLATION COMPARISON TABLE OF NETWORK TOPOLOGY COMPONENT EVOLUTION

Experiment ID	*Algorithm Architecture Components*	*Precision P (%)*	*Recall Rate R(%)*	*mAP50 (%)*	*mAP50−95 (%)*	*Params*	*GFLOPs*
1	Baseline	71.87 ± 1.98	67.67 ± 1.73	73.86 ± 1.27	41.31 ± 1.12	2.37 M	5.2
2	YOLO26-SPD	72.45 ± 1.94	69.60 ± 0.61	74.82 ± 0.57	42.45 ± 0.29	2.47 M	8.4
3	YOLO26-SPD-CA	73.37 ± 1.14	70.24 ± 0.73	74.43 ± 0.24	42.19 ± 0.83	2.94 M	8.7
4	YOLO26-SPD-MSCA	70.53±3.38	70.78±0.56	74.23±1.06	41.97±0.64	2.96M	9.1
5	YOLO26-SPD-CBAM	69.37±3.50	65.58±2.17	71.48±0.90	39.49±1.52	2.94M	8.7
6	YOLO26-SPD-CMSA(Ours)	73.84±1.91	70.12±0.52	75.22±0.55	42.05±0.66	2.94M	8.7

TABLE III. COMPARISON TABLE OF THE RECOGNITION CAPABILITIES OF VARIOUS NETWORK MODULES REGARDING DEFECTS OF DIFFERENT PHYSICAL FORMS

Architecture Group	*Cracking %*	*inclusion %*	*patches %*	*pitted surface %*	*rolled-in scale%*	*scratches %*
Baseline	37.42 ± 2.18	73.65 ± 0.43	92.30 ± 0.57	79.81 ± 2.57	70.93 ± 2.14	89.03 ± 2.01
YOLO26-SPD	39.67 ± 3.16	74.39 ± 0.54	92.41 ± 0.59	82.42± 0.34	68.64 ± 0.34	91.37 ± 1.80
YOLO26-SPD-CA	39.46 ± 1.80	73.91 ± 1.65	92.18 ± 1.45	83.05 ± 1.94	65.41 ± 1.71	92.54 ± 0.94
YOLO26-SPD-MSCA	36.07±1.92	73.19±2.20	92.56±1.54	83.73±0.65	68.31±1.67	91.51±1.73
YOLO26-SPD-CBAM	36.48±1.44	68.34±3.37	92.31±1.30	79.79±3.56	63.52±4.03	88.47±1.18
YOLO26-SPD-CMSA(Ours)	39.22 ±1.10	74.40 ±2.34	91.91 ±0.81	84.87 ±1.93	68.73±3.08	92.18 ±0.97

D. Defect-specific sensitivity analysis

To compare the detection performance of each module on defects with diverse morphologies, Table 3 summarizes per-class mAP50 results repeated three times independently.

The YOLO26-SPD-CMSA achieves optimal performance on most defect categories. For tiny pitted surfaces, it gains 5.06% mAP50 over the baseline and reaches 84.87%, benefiting from the complementary global-local feature extraction of CA and MSCA. For low-contrast crazing cracks, CMSA lifts the mAP50 to 39.22% with the smallest standard deviation, while single MSCA and classic CBAM both lead to obvious accuracy degradation. For long strip scratches, our method obtains 92.18%, with a 3.15% improvement against the baseline. The universal performance decline of CBAM across all defect groups further demonstrates the rationality of the composite CA-MSCA fusion structure.

E. Performance Comparison on the NEU-DET Dataset

An analysis of Table 4 reveals that the proposed YOLO26-SPD-CMSA model demonstrates a balance of low parameter counts and high efficiency in independent testing, as detailed below:

TABLE IV. COMPARISON OF ACCURACY AND MODEL SIZE AGAINST DIFFERENT MODELS ON THE NEU-DET DATASET

Model	*Params (M)*	*mAp50（%）*
YOLOv10n (2024)[16]	2.71	70.00
YOLOv10x (2024)[16]	31.67	72.60
Improving RT-DETR (2025)[17]	10.40	72.90
VTG-YOLOv7-tiny[18]	5.41	74.4
GBS-YOLOv7t (2024)[19]	6.83	72.90
Method in Ref [20](2023)	9.54	73.08
Method in Ref [21](2023)	23.90	74.10
Method in Ref [22]	7.03	77.2
YOLOv8n (2023) [16]	3.01	75.00
YOLO26-SPD-CMSA (Ours)	2.94	75.22

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Experimental results show that the proposed model achieves an excellent mAP50 of 75.22% while maintaining a strictly controlled, lightweight parameter count of 2.94M. When compared against mainstream "heavyweight" networks in the field, the proposed method demonstrates superior energy-efficiency performance.

Specifically, although the heavyweight YOLOv10x [16](introduced in 2024) employs a massive 31.67M parameters, its overall detection accuracy is only 72.60%; in contrast, the proposed model achieves a 2.62% higher accuracy despite a drastically reduced parameter count. Similarly, compared to the latest 2025 improved version of RT-DETR[17], the proposed method achieves higher accuracy while reducing model size. This strongly demonstrates the model's superior spatial utilization during feature extraction, effectively addressing the issue where traditional networks rely on stacking parameters to boost accuracy.

When compared to the VTG-YOLOv7-tiny architecture—which offers comparable accuracy but requires over twice as many parameters—the proposed model cuts the parameter count by nearly 50%[17]. Within the lightweight category specifically designed for edge deployment, cutting-edge solutions such as GBS-YOLOv7 [19]and the method in[20] achieve accuracies of 72.90% and 73.08%, respectively; the proposed model is not only more compact than these alternatives but also delivers consistently higher accuracy. Furthermore, compared to the standard lightweight YOLOv10n[16] of a similar scale, the proposed method achieves a 5.22% accuracy gain with a marginal increase of only 0.23M in parameters. Compared to [22], it reduces the parameter count by more than half while seeing an accuracy drop of only 1.98%; compared to[16], it reduces parameters by 0.07M while improving performance by 0.22%. These results fully validate the advancements in lightweight design achieved by the proposed architecture. In summary, the algorithm proposed in this paper achieves a system-level dynamic balance between model compactness and detection accuracy. While maintaining a lightweight profile, it enhances defect recognition capabilities, offering a valuable lightweight solution for the real-time interception of surface defects on industrial production lines with constrained edge computing resources.

F. R Robustness Verification Against Industrial Domain Shift on GC10-DET

The NEU-DET dataset was collected in a lab with stable lighting, has defects of a single scale, and lacks real steel mill interferences like oxide scales, water stains, rolling patterns, and sensor noise. Relying solely on this dataset can't verify a model's robustness to industrial domain shifts. In this paper, we use the GC10-DET dataset, which has a total of 2,294 real hot-rolled images, to demonstrate robustness to domain changes. The dataset includes lots of complex on-site interferences, making it more aligned with actual production line inspection needs. The training set has 1,605 images, the test set 459, and the validation set 230, covering 10 defect types. The experimental setup is consistent with the above, except that imgsz=1024.

From Table 5, we can see that compared with the original YOLO26, our improved model has a slight drop in precision by 2.93%, but recall goes up by 5.49%, which greatly reduces missed detections of occluded defects. The precision standard deviation drops from 5.88 to 3.90, making predictions more stable under complex noise. mAP50 and mAP50-95 increase by 2.36% and 1.55%, showing that SPD lossless downsampling and CMSA hybrid attention can boost multi-scale defect feature extraction and improve resistance to interference. The steady metric improvements on GC10-DET show that our proposed improvements are robust to real industrial domain shifts, overcoming the limitation of NEU-DET's single scenario.

TABLE V. PERFORMANCE COMPARISON ON GC10-DET

Metric	***Baseline***	***Ours***	***Improvement***
Precision (%)	67.87±5.88	64.94±3.90	-2.93
Recall (%)	58.69±3.75	64.18±1.32	+5.49
mAP50 (%)	63.14±1.59	65.50±1.43	+2.36
mAP50-95 (%)	33.11±0.73	34.66±1.20	+1.55

IV. CONCLUSIONS

Aiming at the limited edge computing resources, low-contrast tiny defects, and high missing detection risks in steel surface inspection, this paper proposes a lightweight YOLO26-SPD-CMSA detection algorithm. By introducing the Space-to-Depth (SPD) module and CMSA hybrid attention mechanism, the network effectively alleviates fine-grained feature loss during downsampling and enhances the feature representation of irregular micro-defects. The proposed method maintains a lightweight scale of 2.94 M parameters and 8.7 GFLOPs, which is friendly for embedded deployment.

All core experiments are repeated three times with results reported as mean ± standard deviation to avoid stochastic training errors and ensure statistical reliability. Quantitative results on NEU-DET show that our method achieves 73.84% Precision, 70.12% Recall, and 75.22% mAP50, surpassing the baseline by 1.97%, 2.45%, and 1.36%, respectively. Controlled ablation experiments confirm that the performance improvement benefits from the structural synergy of CMSA, rather than parameter expansion, while the SPD module only introduces controllable memory overhead.

Furthermore, additional evaluations on the practical GC10-DET dataset verify the model's robustness against real industrial interferences and domain shifts, compensating for the limited scene diversity of the laboratory-controlled NEU-DET dataset. The results demonstrate the good industrial applicability of the proposed lightweight detection method.

Although YOLO26-SPD-CMSA achieves competitive and robust detection performance on both benchmark and industrial datasets, it still has room for improvement. Considering the complexity of actual steel production scenarios, future work will further explore more practical optimization directions for industrial defect detection.

Future plans involve porting the algorithm to resource-constrained edge hardware nodes to conduct round-the-clock,

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real-time, in-line inspection and stability testing in actual workshops where steel strips move at high speeds.

While the current model performs well on defects with distinct contours, there is still room for improvement regarding low-contrast, fine-textured defects. Future work will incorporate additional lightweight modules and consider integrating multimodal data—such as acoustic and infrared information—to further enhance detection rates in harsh operating environments.

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